REAL SPIN BORDISM & ORIENTATIONS OF TOPOLOGICAL K-THEORY

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(joint work with Zach Halladay)

CLASSICAL ORIENTATIONS OF K-THEORY

MAPS OF RING SPECTRA $MG \to E$ $\leftrightsquigarrow$ THOM ISOMORPHISMS IN E-COHOMOLOGY FOR G-VECTOR BUNDLES

$$MU \longrightarrow MSpin^c \longrightarrow KU = (KU_{\mathbb{R}})^e$$

$$MSpin \longrightarrow KO = (KU_{\mathbb{R}})^{C_2}$$

$$MU_{\mathbb{R}} \longrightarrow \boxed{?} \longrightarrow KU_{\mathbb{R}} = KR$$

EVIDENCE FOR $\boxed{?}$ VIA THOM ISOMORPHISMS

C_2 - representation $\mathbb{R}^p \oplus \mathbb{R}^q\sigma$

Conjugate linear $C_2 \circlearrowright \mathbb{C}\ell(\mathbb{R}^{p,q})$

$$\begin{gathered} \mathbb{C}\ell(\mathbb{R}^{p,q}) \\ \cup\!\!\shortmid \\ \Rightarrow \quad C_2 \circlearrowright \mathrm{Spin}^c(p,q) \end{gathered}$$

Theorem (Atiyah): $G \longrightarrow \mathrm{Spin}^c(p,q)$ C_2-equivariant

$\Rightarrow KU_{\mathbb{R},G}(X) \xrightarrow[\cdot u]{\sim} KU_{\mathbb{R},G}(\mathbb{R}^{p,q} \times X)$, $p \equiv q \mod 8$.

↳ contains all Thom isomorphisms above,

e.g. $\mathrm{Spin}(8n) \hookrightarrow \mathrm{Spin}^c(8n) = \mathrm{Spin}^c(8n,0)$

& $U(n) \hookrightarrow \mathrm{Spin}^c(n,n)$ C_2-equivariantly

Main Theorem

$\exists$ C_2-E_∞-ring spectrum $MSpin^c_{\mathbb{R}}$ such that

(1) $MSpin^c = (MSpin^c_{\mathbb{R}})^e$

(2) $\exists$ E_∞-map $MSpin \longrightarrow (MSpin^c_{\mathbb{R}})^{C_2}$

(3) $\exists$ C_2-E_∞-maps $MU_{\mathbb{R}} \longrightarrow MSpin^c_{\mathbb{R}} \longrightarrow KU_{\mathbb{R}}$

Real spin orientation

(1)+(3) $\Longrightarrow$ $MSpin^c \to KU$

(2)+(3) $\Longrightarrow$ $MSpin \longrightarrow KO$

(3) $\Longrightarrow$ $MU_{\mathbb{R}} \to KU_{\mathbb{R}}$

THE FIXED POINTS OF $MSpin^c_{\mathbb{R}}$

<u>Recall:</u>

$$\begin{tikzcd}
Spin(n) \arrow[rr] \arrow[dr, "\exists"'] & & Spin^c(n) \\
& Spin^c(n)^{C_2} \arrow[ur, hook] &
\end{tikzcd}$$

<u>By (2):</u>

$$\begin{tikzcd}
MSpin \arrow[rr, "u"] \arrow[dr, "\exists w"'] & & MSpin^c \\
& (MSpin^c_{\mathbb{R}})^{C_2} \arrow[ur, "res"'] &
\end{tikzcd}$$

<u>Theorem:</u>

Any w factoring u is not an equivalence.

Proof: Suppose w is an equivalence.

$\Rightarrow$ Mackey functor

$$\begin{tikzcd} MSpin_* \arrow[r, bend left, "u"] & MSpin^c_* \arrow[l, bend left, "tr"] \arrow[loop right, "C_2 \ni \overline{(\ )}"] \end{tikzcd}$$

$$\Rightarrow \alpha + \bar{\alpha} = u(tr(\alpha)) \in u(MSpin_*), \quad \alpha \in MSpin^c_*$$

$$\Rightarrow \boxed{\hat{A}(\alpha + \bar{\alpha}) \in \mathbb{Z}, \quad \forall \alpha \in MSpin^c_* \quad (\star)}.$$

$$\hat{A}([\mathbb{CP}^2]) = -\frac{1}{8} \Rightarrow \hat{A}([\overline{\mathbb{CP}^2}]) = \frac{1}{8} + n$$

$$\Rightarrow \hat{A}([\mathbb{CP}^2]^2 + [\overline{\mathbb{CP}^2}]^2) = \frac{8n+1}{32} + n^2 \notin \mathbb{Z} \; !$$

$\hookrightarrow$ Actually shows $(MSpin \xrightarrow{u} MSpin^c) \not\simeq (E^{C_2} \to E^e)$ $\forall E$

THANKS FOR LISTENING!

$\boxed{\text{OVERVIEW}}$

$$\begin{tikzcd}
 & MSpin \arrow[r] \arrow[d, "\not\simeq"] & KO \arrow[d, "\simeq"] \\
 & (MSpin^c_{\mathbb{R}})^{C_2} \arrow[r] \arrow[d] & (KU_{\mathbb{R}})^{C_2} \arrow[d] \\
MU_{\mathbb{R}} \arrow[r] & MSpin^c_{\mathbb{R}} \arrow[r] & KU_{\mathbb{R}}
\end{tikzcd}$$